INFOSYS SPRINGBOARD

Virtual Internship Project Report

TOPIC:

Multimodal AI System for Automated Health Diagnostics

(Blood Report Analysis & Personalized Recommendations)

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Abstract:

The rapid growth of digital healthcare data has created a need for intelligent systems capable of interpreting medical information efficiently. Blood reports provide crucial insights into a person's health status, but the complexity of medical terminology often makes them difficult for non-experts to understand.

This project presents a **Multimodal AI System for Automated Health Diagnostics** that analyzes blood reports using multiple AI models to detect abnormalities, identify health risks, and generate personalized recommendations. The system integrates data extraction, parameter interpretation, pattern recognition, and contextual analysis into a unified pipeline. The proposed solution aims to improve accessibility to healthcare insights and support early detection of potential health issues while emphasizing that it is not a substitute for professional medical advice.

Introduction:

Advancements in artificial intelligence have enabled the development of systems capable of assisting in medical decision support. Blood tests are among the most commonly used diagnostic tools, providing valuable information about organ function, nutritional status, and disease risk. However, interpreting these reports requires medical expertise, and patients often struggle to understand the implications of their results.During my **Infosys Springboard Virtual Internship**, I developed an AI-based diagnostic assistant that automates the interpretation of blood reports. The system leverages multiple AI models to provide accurate analysis and meaningful insights. By converting complex medical data into simple summaries and recommendations, the project demonstrates the potential of AI to enhance preventive healthcare and patient awareness.

Project Overview:

The system is a multimodal AI platform that processes blood reports in digital formats and performs automated analysis. It combines parameter interpretation, pattern recognition, and contextual analysis to produce reliable insights.

The final output is a user-friendly diagnostic summary highlighting abnormal parameters, possible health risks, and personalized recommendations.

Objectives:

- Automate extraction of blood parameters from reports
- Analyze values against medical reference ranges
- Detect correlations indicating potential health risks
- Generate personalized recommendations
- Provide a simple and clear diagnostic summary

Problem Statement:

Blood reports contain numerous medical parameters that require expert interpretation. Patients without medical knowledge may find it difficult to understand these reports, which can lead to confusion or delayed medical action.

An automated system can help bridge this gap by providing quick, understandable insights.

System Workflow:

1. User uploads blood report (PDF/Image/json/csv).
2. OCR extracts parameters, values, and units.

3. Data is cleaned and standardized.

4. Multiple AI models analyze the data.

5. Findings are combined into a summary.

6. Personalized recommendations are generated.

7. Final report is displayed to the user.

Multimodal AI Models:

Parameter Interpretation Model: Compares each parameter with standard medical ranges and classifies values.

Pattern Recognition Model: Identifies relationships between parameters and calculates risk indicators.

Contextual Analysis Model: Uses user details such as age or gender to refine results and provide personalized insights.

System Architecture:

The system follows a modular architecture consisting of input layer, data processing module, AI analysis models, recommendation engine, and report generation module. This ensures scalability and flexibility.

Milestone 1 – Data Ingestion

Implemented report upload functionality and OCR-based parameter extraction.

Data sources and reference ranges were taken from WHO guidelines, CDC resources, and sample datasets.

Milestone 2 – Pattern Recognition

Developed AI model to analyze parameter relationships and detect risk patterns.

Milestone 3 – Recommendation Engine

Generated personalized health suggestions based on identified abnormalities.

Milestone 4 – Integration

Integrated all modules into a single pipeline and produced final diagnostic reports.

Tools and Technologies

Python, Machine Learning libraries, OCR tools, NLP/LLM models, Streamlit/Tkinter, Pandas, NumPy.

Results

The system successfully extracts parameters, detects abnormalities, and generates meaningful insights. The multimodal approach improves analysis accuracy and reliability.

Advantages and Limitations

Advantages: Fast analysis, easy interpretation, supports preventive healthcare.

Limitations: Not a substitute for professional diagnosis, depends on report quality.

Conclusion

The project demonstrates the potential of multimodal AI in healthcare diagnostics by automating blood report interpretation and providing actionable insights.